

RESEARCH STATEMENT

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*When the Model Becomes the Risk: A Research Program in Scalable
Bayesian Inference for Trustworthy Environmental Health Surveillance*

My research builds Bayesian statistical methods for environmental and health data collected across many locations over many years. The main question that runs through my work is simple: how do we build large hierarchical models that scale to national surveillance data while still giving us trustworthy scientific conclusions? I study this question using Bayesian statistics, spatial statistics, numerical linear algebra, and dimension reduction. I develop these methods together with the environmental health problems that motivate them. My goal as an independent researcher is to build a long-term, externally funded research program, and to train the students who will carry this work forward. The program sits at the meeting point of scalable Bayesian methodology and environmental health surveillance.

Research Vision and Significance

Federal and public health agencies study environmental risk using data from thousands of locations and many years. Every regulatory decision based on these data depends on the statistical model used to separate the effect of an exposure from the spatial and temporal patterns around it. Hierarchical Bayesian models are the standard tool for this work. At national scale, however, these models often fail in one specific way. When we combine spatial and temporal random effects, the model produces a very large set of near-duplicate components. This slows down the computation. More importantly, the random effects can absorb signal that should belong to the exposure of interest. In the worst cases, the model reverses the apparent direction of the effect, so that a harmful exposure looks protective. When a modeling artifact can flip the sign of a regulatory conclusion, the method itself becomes a public health risk.

The value of my research program is that it treats this as a foundational problem rather than something to patch one analysis at a time. My main contribution, SVD-PICAR, shows that scale and reliability do not need to be traded against each other. SVD-PICAR uses a singular value decomposition of the projected intrinsic conditional autoregressive basis. This reduces a spatio-temporal model to the few orthogonal directions that actually explain the outcome. The model then learns its own dimension from the data rather than relying on a guess from the analyst. The work has a clear audience. Agencies such as NIEHS, CDC-NCHS, NCI, and EPA all use exactly these kinds of models. This places the program in a strong position to attract long-term external funding from NIH R01, NSF DMS, and EPA mechanisms, while still producing methodology of independent statistical interest.

Foundational Contributions

My dissertation provides the foundation that my independent program will build on. I summarize the two main contributions below.

1. *Compressed spatio-temporal inference for Gaussian and multinomial outcomes.*

When spatial and temporal structure are combined, the result is a large set of near-duplicate components that slow down computation and make inference unstable. I showed how the singular value decomposition gives the best low-rank summary of this structure. I then used adversarial simulations to show that the compressed model keeps both accuracy and well-calibrated uncertainty, while running with the speed and stability of a much smaller model [1]. I validated the method on large real datasets, including NASA satellite environmental measurements and national cancer surveillance records.

2. *Confounding-aware mixture effects for over-dispersed count data.*

Bayesian kernel machine regression is the standard method for estimating the joint effect of correlated chemical exposures. The standard form, however, does not handle joint spatial and temporal dependence in count outcomes. When this dependence is ignored, the estimated mixture effect can be biased and can even reverse direction. My SVD-PICAR negative-binomial BKMR cleanly separates the spatio-temporal background from the chemical signal. Applied to county-level thyroid cancer counts and Nebraska pesticide records, the method correctly identified the chemicals linked to elevated cancer risk. These results match established toxicological evidence and could not be recovered using the standard approach [2]. The same foundation also supported a collaborative epidemiologic study linking drought to COPD mortality [3], which shows the program can produce cross-disciplinary, co-authored science.

An Independent Research Program: Three Funded Thrusts

My program moves forward along three thrusts. Each thrust is a self-contained research direction that can support a federal grant proposal and a PhD student's dissertation.

1. *Spatially varying exposure effects (anchor for an NIH R01).* A pesticide does not have the same health effect everywhere. Local conditions such as application intensity, hydrology, and co-exposure profiles all matter. I will place a spatially varying coefficient process inside the compressed basis. I will then derive the SVD-PICAR

version of restricted spatial regression, so that the exposure surface is estimated free of spatial confounding by construction. The methodological core is the identifiability problem between a flexible coefficient surface and a flexible confounder surface. This is interesting in its own right. The applied output, county level effect maps with calibrated uncertainty, is directly fundable through environmental health mechanisms. This thrust will support one to two PhD students.

2. *Critical exposure windows (anchor for an NSF DMS proposal)*. Setting an evidence-based exposure limit requires knowing not only whether an exposure is harmful but also *when* it does harm. I will develop a distributed-lag method inside the SVD-PICAR framework. The temporal SVD directions will play two roles at once: they will capture the background temporal trend and span a penalized low-rank surface for the lag effects. I will add a fused penalty so that the lag effects are smooth across adjacent years, as toxicology suggests, while still allowing a sharp critical window to appear when the data support one. The theoretical questions, such as the rank behavior of the joint trend-lag basis and the conditions under which the window is identifiable, make this a natural NSF methodology proposal and a strong dissertation problem.
3. *Automatic model selection and reproducible software (broader impact and sustainability)*. The main barrier between a good method and its routine use is the expert judgment the method silently asks for. I will develop and study three model selection procedures: WAIC, leave-one-out cross-validation, and a marginal likelihood criterion based on the SVD basis. I will establish consistency guarantees and then release an open-source pipeline that an agency analyst can run without needing a statistician to make modeling choices at each step. The software is both a broader-impacts component for grant proposals and the main mechanism by which the program's methods will reach practitioners and accrue citations.

Mentoring, Teaching, and Departmental Synergy

I see research and training as one activity, not two. Each thrust above is scoped so that students enter at a clean, tractable subproblem and graduate with a complete methodological contribution. My experience as a graduate research assistant in the UNMC Water, Climate and Health Program, and as a supplementary instructor and teaching assistant across statistics and mathematics, has shaped a mentoring approach built on three things: giving students early ownership of their work, providing frequent feedback, and always pairing a method with the applied problem that gives it meaning. I am ready to teach the core graduate biostatistics courses, including Bayesian methods, hierarchical and mixed models, categorical data analysis, and statistical computing in R and Python. I would also like to develop an advanced elective in spatio-temporal modeling that brings students into the program's thrusts.

My program is built for synergy. It produces methodology that collaborators in environmental health, epidemiology, and spatial statistics can pick up and use. In return, the program benefits from access to surveillance data and substantive expertise. My NASA ARSET certifications in satellite-derived environmental data put me in a strong position to bridge statistical groups and remote-sensing groups. My co-authored, cross-disciplinary work shows that the program produces methods of independent interest and also embeds those methods in real collaborative science.

Long-Term Trajectory

In the near term, my program produces methodology with a clear audience and clear funding pathways. In the longer term, as surveillance systems get larger and more interconnected, three issues will grow in importance alongside scalability: robustness to model misspecification, graceful degradation when data quality is uneven, and reproducibility of results. Dimension reduction will remain the organizing principle that keeps these problems tractable. My goal is to build, at an R1 institution, a long-term and externally funded research group known for two things: methodology that is statistically rigorous, and methodology that directly helps protect public health. I am also committed to training the next generation of students to extend this work. I am excited by the prospect of contributing to this growing field and making a lasting impact in it.

References

- [1] K. B. Afari, Y. Gwon, "Compressed Spatio-Temporal Basis Functions: An SVD-PICAR Framework for Scalable Gaussian and Multinomial Inference," submitted to *Journal of Computational and Graphical Statistics*, 2026. Under review.
- [2] K. B. Afari, Y. Gwon, "SVD-PICAR Bayesian Kernel Machine Regression Framework for Spatio-Temporal Count Data," submitted to *Statistics in Medicine*, 2026. Under review.
- [3] K. C. Gribben, Y. Gwon, K. Afari, J. Berman, A. Li, B. Fard, et al., "Drought increases the risk of COPD mortality in the United States," submitted to *Environmental Research*, 2026. Under review.